Jonathan Ung

Canadian Citizen | US J-1 Eligible
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Education

Simon Fraser University - Bachelor of Science in Computer Science **EXPECTED APR 2028**

Relevant Coursework: Data Structures & Algorithms, Systems Programming, Distributed Systems, Artificial Intelligence, Networking, Computer Graphics, Computer Vision, Data Science, Robotic Autonomy

Experience

Qualcomm | Markham, ON, Canada **MAY 2026 – AUG 2026**
AI Infrastructure Software Engineering Intern *C/C++, Python, ONNX, GenAI, QNN, Hexagon*

- Enabled GenAI models exceeding device RAM to run on-device through the **Genie SDK** while minimizing inference latency by designing an **MRU-based graph eviction strategy** which dynamically pages KV-cache graph partitions
- Built automated model validation tool for the **Hexagon NPU** graph pipeline, detecting **8+ categories** of ONNX-to-QNN graph compilation defects including quantization mismatches, redundant data movers, and broken connectivity
- Integrated the graph validator into **Jenkins CI** for model-zoo submissions, created an on-demand validation agent for pre-submission models, and exposed an API the **QAIRT SDK**, enabling model diagnostics across internal and vendor workflows
- Unified **70+ HexNN math and approximation functions** into an accessible library backed by rigorous unit and on-device testing integrated into **Jenkins CI** for automated regression coverage

Rivian and Volkswagen Group Technologies | Vancouver, BC, Canada **MAY 2025 – DEC 2025**
Infotainment Software Engineering Intern *Kotlin, Java, Android, Protobuf, QNX*

- Owned **12+ user-facing features** across Rivian R2 Media App (browsing, search, playback, shuffle, podcasts), delivering an audio visualizer and vendor integrations for Spotify, Apple Music, and Tidal to pre-production builds
- Migrated Media App from fragment-based architecture to MVVM with view controllers consuming data models, cutting RAM usage **40%** and accelerating feature velocity **30%**
- Resolved critical vehicle safety defects including cold-boot launch crashes, unsafe service data handling, and a radio refactor regression that escaped QA, collaborating across platform and QA teams
- Fixed Bluetooth AVRCP metadata errors via caching and content-hash validation, raising accuracy **from 66% to 99%**

SFU Robot Soccer | Burnaby, BC, Canada **FEB 2024 – DEC 2026**
Software Team Lead *C++, Python, Qt6, Protobuf, Docker*

- Led **30-member** C++ team; redesigned onboarding workflows reducing new developer ramp-up time **40%**
- Engineered Behavior Tree DSL reducing node boilerplate **10:1** and Play Graph state machine with 12-play transitions evaluating **50+ metrics/tick**, coordinating 6 robots at **180Hz**
- Designed Protobuf networking protocol routing commands to **6 robots** with hot-swappable simulator/hardware modes

Projects

VeTool - FPS Matchmaking Service *.NET, PostgreSQL, Next.js, SignalR, Redis, Docker*

- Built full-stack matchmaking platform for FPS custom lobbies with real-time drafting, pick/ban, and match history tracking, serving **50+ active users**
- Delivered SignalR and Redis event layer achieving **sub-150ms** state sync across horizontally scaled API instances
- Designed high-concurrency PostgreSQL schema with transactional safeguards and indexed lookups for conflict-free writes

Nasdaq ITCH Aggregator - GPU-Accelerated Market Data Pipeline *C++, CUDA, CMake, Python, zlib*

- Built GPU-accelerated Nasdaq ITCH 5.0 pipeline streaming full-day TotalView feeds through a CPU parser into CUDA kernels, computing per-symbol **VWAP**, order imbalance, and bid/ask spread across **16,384 symbols**
- Overlapped CPU parsing, pinned-memory transfers, and **CUDA** execution using a four-slot asynchronous ring buffer, enabling continuous high-throughput processing without blocking the ingest path

Agentic Code Harness Tooling *Python, Go, Docker, Shell, tmux, Claude Code Plugin API*

- Developed **Zone** under **Peasant Labs**, a per-repo Docker workspace manager for LLM coding agents with network sandboxing, adopted by **10+ researchers** for safe autonomous code execution
- Created **Finesse, a wave-based task decomposition** with dependency-aware scheduling, automatically splitting large tasks into parallel sub-workflows across isolated git worktrees with tmux orchestration and merge reconciliation

Technical Skills

Languages: Python, C/C++, CUDA, C#, Java, Kotlin, JavaScript, TypeScript, Go, SQL, Bash
Frameworks: React, Next.js, .NET, FastAPI, Node.js, Express.js, Spring Boot, Electron, SignalR, GraphQL
Tools & Infrastructure: Docker, AWS (EC2, S3, Lambda), PostgreSQL, MongoDB, Redis, Git, GitHub & GitLab CI/CD, Linux
Concepts: API Design, Distributed Systems, Developer Tooling, Real-Time Systems, LLMs, Applied Machine Learning, Robotics